DSA Day 15: Expression Conversion (infix->postfix)

Book: Data Structures Through C in Depth - Ch4 §4.6 · pp 139-146 | Code in Python

Overview

Use a stack to convert infix to postfix respecting precedence and associativity, then evaluate postfix with a stack.

Key concepts to master

- Operator precedence & associativity
- Stack of operators
- Evaluate postfix

Python implementation

```
prec={'+':1,'-':1,'*':2,'/':2}
def to_postfix(e):
    out=[]; st=[]
    for t in e:
        if t.isalnum(): out.append(t)
        elif t=='(' : st.append(t)
        elif t==')':
            while st and st[-1]!='(': out.append(st.pop())
            st.pop()
        else:
            while st and st[-1]!='(' and prec.get(st[-1],0)>=prec[t]: out.append(st.pop())
            st.append(t)
    return ''.join(out+st[::-1])
```

Complexity

O(n).

Common pitfalls

- Mishandling parentheses
- Wrong precedence for same-level operators

Practice

- Convert (a+b)*(c-d)
- Evaluate 2 3 + 4 *